One load, many idle rods

Plane truss — analysis and CIRSOC 301 rod checks

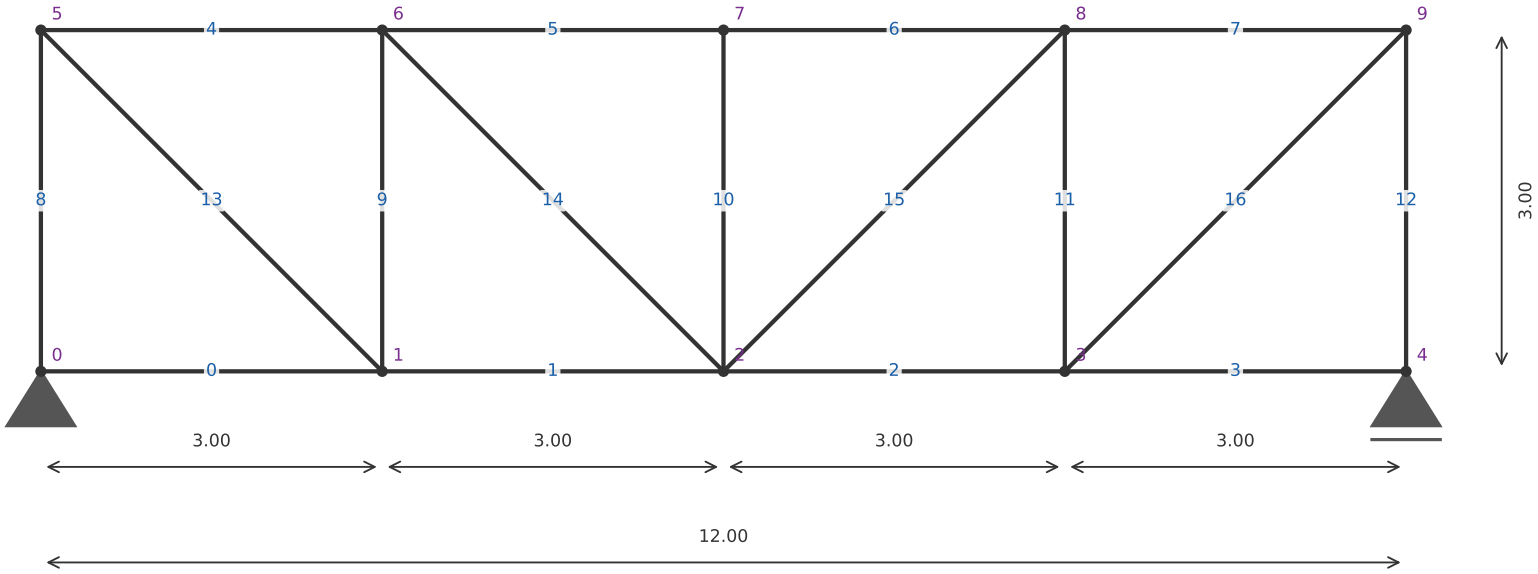
Span	12.00 m
Height	3.00 m
Nodes / rods	10 / 17
Supports	pin at 0, rollerX at 4
Loads	1 nodal, total 60.0 kN
Family "Chord"	CHS 114.3x4.5 (F24 (CIRSOC) / A36)
Family "Diagonal"	CHS 60.3x3.6 (F24 (CIRSOC) / A36)
Family "Vertical"	CHS 88.9x4.0 (F24 (CIRSOC) / A36)
Weight	Weighs 513 kg, carries 60.0 kN (12 × its own weight), passes (0.95).
Deflection	4.62 mm = L/2595, limit L/300: OK
Stability	Statically determinate (m + r − 2j = 17 + 3 − 2×10 = 0): statics alone gives every force.
Most used rod	rod 14 at 0.95 (elastic buckling (E3-3))
Rods over capacity	0 of 17 checked
Buckling	$\lambda_1 = 1.35$
Shear and moment	none: every rod is pin-ended and loaded only at its nodes, so the axial force is the whole result

Contents

2. Geometry
3. Axial forces
4. Deflection and stress
5. Reactions
6. Utilisation
7. Buckling
8. Rods
9. Explain rod 14 and 13

Results are an aid to a qualified professional who reviews them. Rods modelled as pin-ended members; CIRSOC 301 assumptions; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



2 m · true to scale

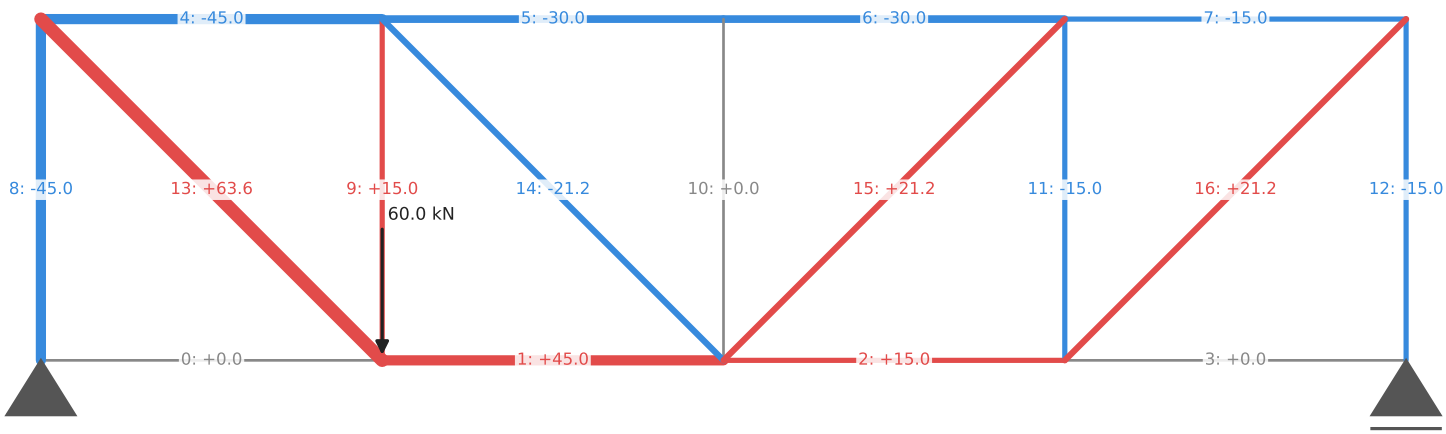
0 2

Truss — Geometry

Sheet 2 / 9

MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 63.6).



2 m · true to scale

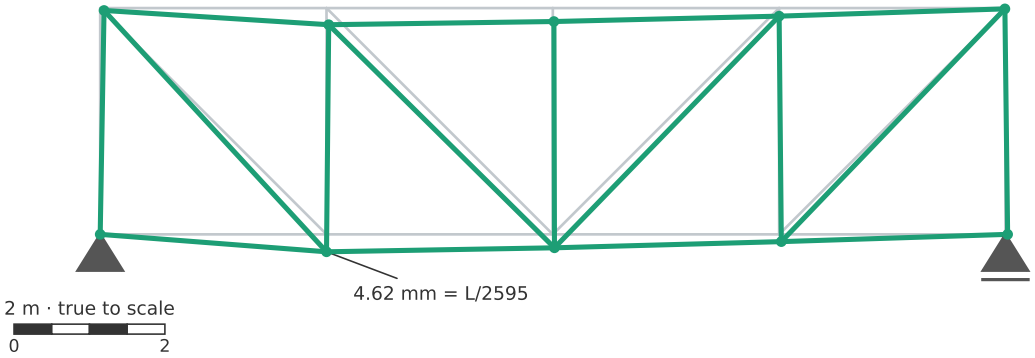
0 2

Truss — Axial forces

Sheet 3 / 9

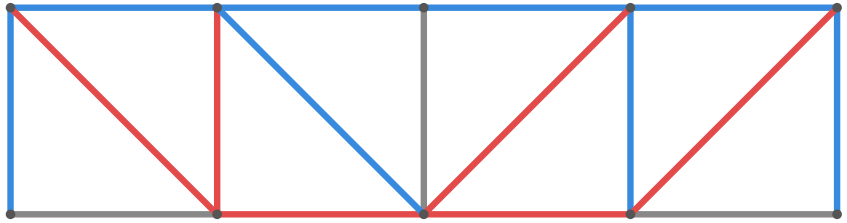
MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50.
Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 4.62 mm against the limit L/300 = 40.0 mm: OK.

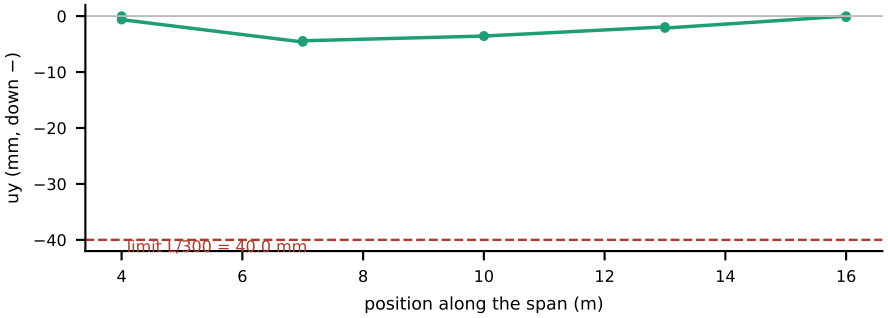


Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

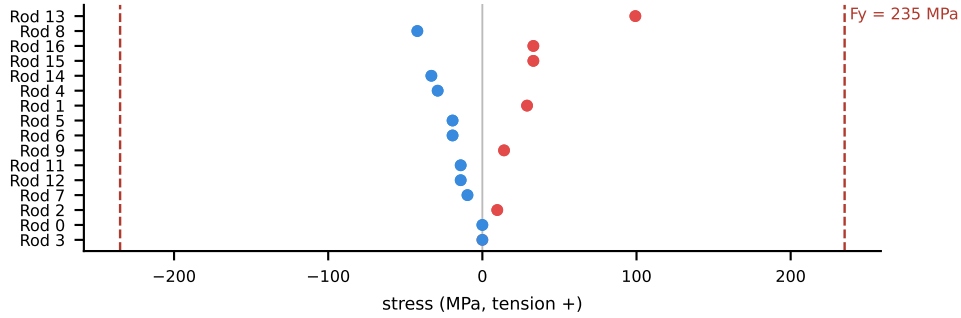


Deflection along the span vertical movement of every joint



Largest 4.62 mm at joint 1 = L/2595. Within the limit.

Fibre stress values the two faces of each rod, MPa



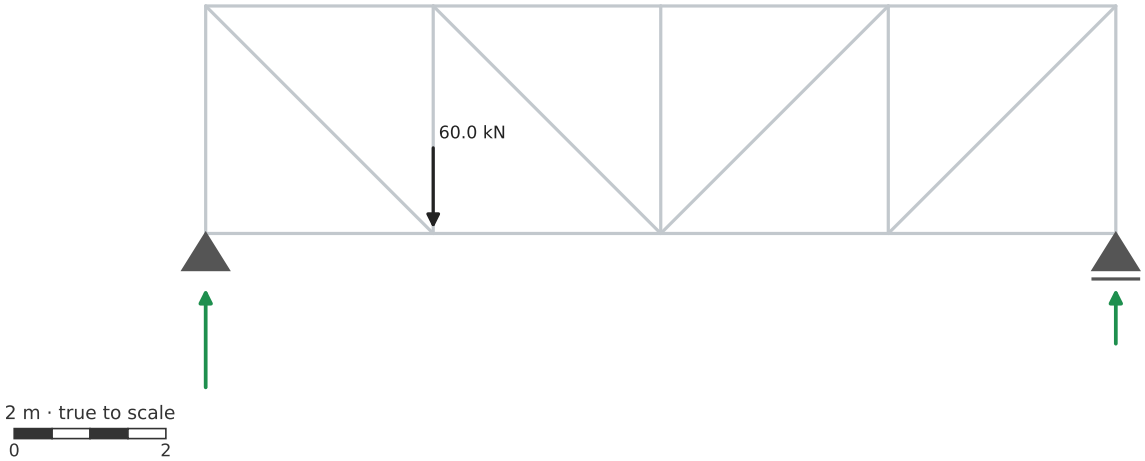
Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

Truss — Deflection and stress

MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

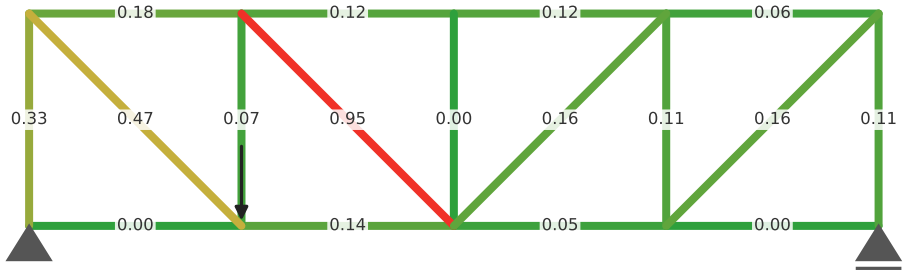
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	45.00
4	rollerX	0.00	15.00
Σ		0.00	60.00



Truss — Reactions

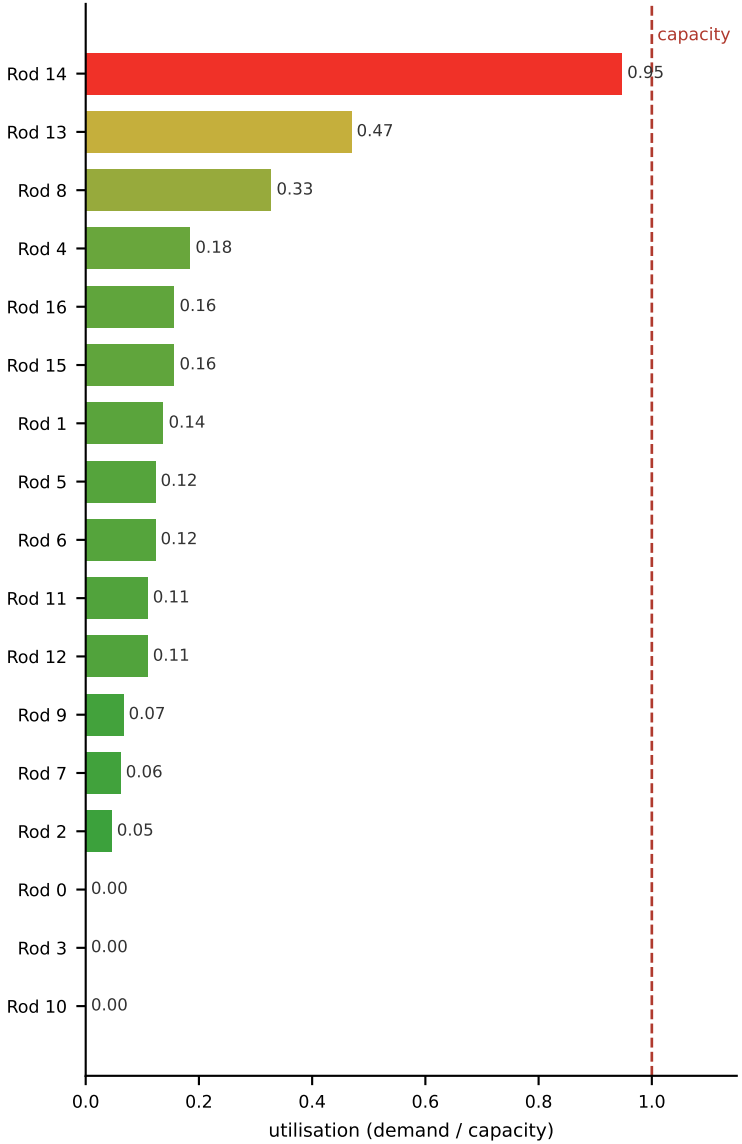
MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

Utilisation: demand over CIRSOC 301 capacity. Most used rod 14 at 0.95 (elastic buckling (E3-3)). Every checked rod is within capacity.



2 m · true to scale

0 2





grey = no steel section, not checked

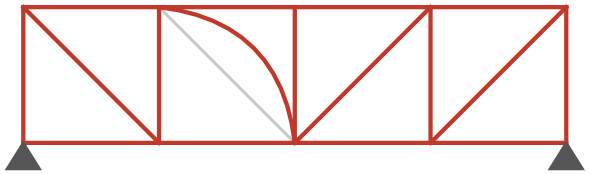
Truss — Utilisation

Sheet 6 / 9

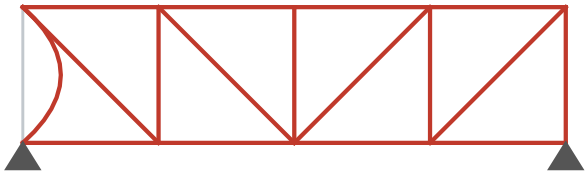
MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

λ is the factor on the load as it stands. λ_1 = 1.35, so the load can grow 1.3× before it buckles. Mode 2 is 4.73, 251 % higher: one member decides the buckling load.

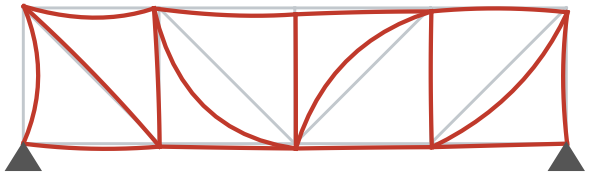
Mode 1: λ = 1.35
rod 14 bows

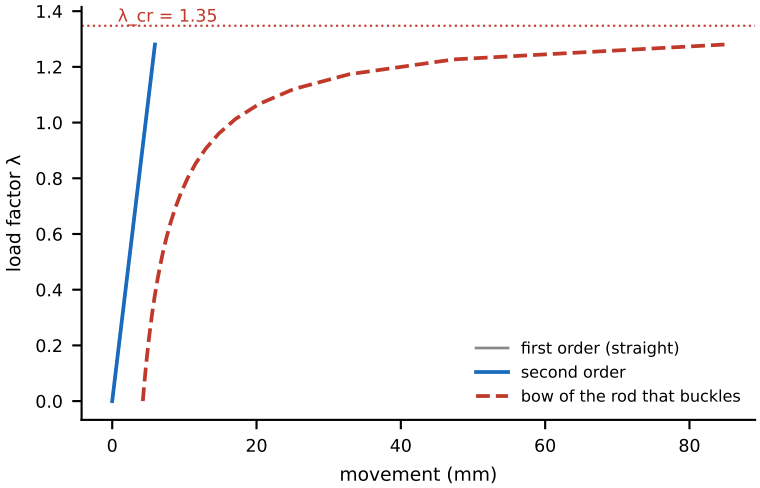


Mode 2: λ = 4.73
rod 8 bows



Mode 3: λ = 6.51
rod 14 bows





The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.24 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	0–1	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00	OK
1	1–2	Chord	CHS 114.3x4.5	3.00	+45.00	normal stress (H.3.4)	0.14	OK
2	2–3	Chord	CHS 114.3x4.5	3.00	+15.00	normal stress (H.3.4)	0.05	OK
3	3–4	Chord	CHS 114.3x4.5	3.00	+0.00	normal stress (H.3.4)	0.00	OK
4	5–6	Chord	CHS 114.3x4.5	3.00	-45.00	inelastic buckling (E3-2)	0.18	OK
5	6–7	Chord	CHS 114.3x4.5	3.00	-30.00	inelastic buckling (E3-2)	0.12	OK
6	7–8	Chord	CHS 114.3x4.5	3.00	-30.00	inelastic buckling (E3-2)	0.12	OK
7	8–9	Chord	CHS 114.3x4.5	3.00	-15.00	inelastic buckling (E3-2)	0.06	OK
8	0–5	Vertical	CHS 88.9x4.0	3.00	-45.00	inelastic buckling (E3-2)	0.33	OK
9	1–6	Vertical	CHS 88.9x4.0	3.00	+15.00	normal stress (H.3.4)	0.07	OK
10	2–7	Vertical	CHS 88.9x4.0	3.00	+0.00	normal stress (H.3.4)	0.00	OK
11	3–8	Vertical	CHS 88.9x4.0	3.00	-15.00	inelastic buckling (E3-2)	0.11	OK
12	4–9	Vertical	CHS 88.9x4.0	3.00	-15.00	inelastic buckling (E3-2)	0.11	OK
13	5–1	Diagonal	CHS 60.3x3.6	4.24	+63.64	normal stress (H.3.4)	0.47	OK
14	6–2	Diagonal	CHS 60.3x3.6	4.24	-21.21	elastic buckling (E3-3)	0.95	OK
15	8–2	Diagonal	CHS 60.3x3.6	4.24	+21.21	normal stress (H.3.4)	0.16	OK
16	9–3	Diagonal	CHS 60.3x3.6	4.24	+21.21	normal stress (H.3.4)	0.16	OK



Rod 14

utilisation 0.95

elastic buckling (E3-3)

```
Rod 14
======
Steel, CIRSOC 301-2018 -- A = 641 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 20.1 mm
      = 211.2

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 211.2²
      = 44.25 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 211.2 vs 4.71 √(200,000/235) = 137.4
      = elastic

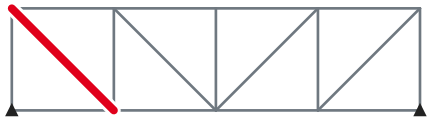
 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 44.25
      = 38.80 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 38.80 × 641
      = 22.40 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 21.21 / 22.40
      = 0.947

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 0.947

utilisation 0.947 -- OK
```



Rod 13

utilisation 0.47

normal stress (H.3.4)

```
Rod 13
======
Steel, CIRSOC 301-2018 -- A = 641 mm², Fy = 235 MPa, L = 4.243 m

 1. Axial stress
      σ = N / A
      = 63,640 N / 641 mm²
      = 99.24 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 99.24 / 211.50
      = 0.469

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.469

utilisation 0.469 -- OK
```

Truss — Explain rod 14 and 13

MODEL	SIZE	UNITS	CODE	DATE
One load, many idle ro	10 nodes / 17 rods	m, kN, mm	CIRSOC 301	2026-10-12 12:05